

Problem Set: INDE301__Ch1__Notes__24.pdf

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Chapter Analysis

Topics Covered: - Engineering Economy - Decision-Making Process - Economic Analysis in Engineering

Key Formulas:

No formulas identified

Problem 1

Difficulty: EASY

Topic: Decision-Making Process

Problem Statement

List the first three steps in the decision-making process as outlined in the chapter.

Find: - List the first three steps in the decision-making process.

Solution

1. Approach & Strategy

- Identify the first three steps in the decision-making process as outlined in the chapter.
- Present these steps in a clear and concise format.

2. Step-by-step Calculations

Step 1: Understand the Problem and Define the Objective

- This step involves recognizing the issue at hand and clarifying what needs to be achieved.
- **Given:**
 - Problem recognition
 - Objective definition

Step 2: Collect Relevant Information

- In this step, gather all necessary data and information that will inform the decision-making process.
- **Given:**
 - Data collection
 - Information relevance

Step 3: Define Alternatives and Estimate Relevant Costs

- This step requires identifying different options available and assessing the costs associated with each alternative.
- **Given:**
 - Alternatives identification
 - Cost estimation

3. Final Answer

- **Answer:** The first three steps in the decision-making process are:
 1. Understand the problem and define the objective.
 2. Collect relevant information.
 3. Define alternatives and estimate relevant costs.
- **Check:** These steps are fundamental to effective decision-making in engineering economy.

Quality Assessment: good

Issues: - The problem statement is clear, but it lacks context or relevance to a specific scenario, which could enhance engagement. - The solution includes unnecessary sections like ‘Approach & Strategy’ and ‘Step-by-step Calculations’ that do not directly contribute to answering the problem. - The use of ‘Given’ in the solution is misleading as it does not pertain to the problem context; it should be omitted.

Problem 2

Difficulty: MEDIUM

Topic: Estimating financial outcomes

Problem Statement

A company is considering two alternatives for a new project. Alternative A has an estimated cost of \$150,000 and is expected to generate \$200,000 in revenue. Alternative B has an estimated cost of \$120,000 with expected revenue of \$180,000. Calculate the net financial outcome for each alternative.

Given: - Cost of Alternative A: \$150,000 - Revenue from Alternative A: \$200,000 - Cost of Alternative B: \$120,000 - Revenue from Alternative B: \$180,000

Find: - Net financial outcome for Alternative A - Net financial outcome for Alternative B

Solution

1. Approach & Strategy

- Calculate the net financial outcome for each alternative by subtracting the estimated cost from the expected revenue.
- Compare the results to determine which alternative is financially more favorable.

2. Step-by-step Calculations

Step 1: Calculate Net Financial Outcome for Alternative A

- The net financial outcome is calculated using the formula:

$$\text{Net Outcome} = \text{Revenue} - \text{Cost}$$

- **Given values:**
 - **Cost of Alternative A:** \$150,000
 - **Revenue from Alternative A:** \$200,000

$$\text{Net Outcome A} = 200,000 - 150,000$$

Step 2: Calculate Net Financial Outcome for Alternative B

- Using the same formula for Alternative B:

$$\text{Net Outcome} = \text{Revenue} - \text{Cost}$$

- **Given values:**
 - **Cost of Alternative B:** \$120,000
 - **Revenue from Alternative B:** \$180,000

$$\text{Net Outcome B} = 180,000 - 120,000$$

3. Final Answer

- **Answer:**
 - Net financial outcome for Alternative A: **\$50,000**
 - Net financial outcome for Alternative B: **\$60,000**

- **Check:** Both outcomes are positive, indicating that both alternatives are financially viable, with Alternative B being the more favorable option.

Quality Assessment: excellent

Problem 3

Difficulty: HARD

Topic: Evaluating costs

Problem Statement

A company has three alternatives for upgrading its manufacturing process. The estimated costs and expected savings over five years are as follows: Alternative X costs \$500,000 and saves \$600,000; Alternative Y costs \$450,000 and saves \$550,000; Alternative Z costs \$400,000 and saves \$450,000. Evaluate each alternative based on the net savings and determine which alternative is the best choice.

Given: - Cost of Alternative X: \$500,000, Savings: \$600,000 - Cost of Alternative Y: \$450,000, Savings: \$550,000 - Cost of Alternative Z: \$400,000, Savings: \$450,000

Find: - Net savings for Alternative X - Net savings for Alternative Y - Net savings for Alternative Z - Best alternative based on net savings

Solution

1. Approach & Strategy

- Calculate the net savings for each alternative by subtracting the cost from the savings.
- Compare the net savings of all alternatives to determine the best choice.

2. Step-by-step Calculations

Step 1: Calculate Net Savings for Alternative X

- To find the net savings for Alternative X, we will use the formula:

$$\text{Net Savings} = \text{Savings} - \text{Cost}$$

- **Given values:**
 - **Cost of Alternative X:** \$500,000
 - **Savings of Alternative X:** \$600,000

$$\text{Net Savings for X} = 600,000 - 500,000 = \mathbf{100,000}$$

Step 2: Calculate Net Savings for Alternative Y

- Now, we will calculate the net savings for Alternative Y using the same formula.
- **Given values:**
 - **Cost of Alternative Y:** \$450,000
 - **Savings of Alternative Y:** \$550,000

$$\text{Net Savings for Y} = 550,000 - 450,000 = \mathbf{100,000}$$

Step 3: Calculate Net Savings for Alternative Z

- Next, we will calculate the net savings for Alternative Z.
- **Given values:**
 - **Cost of Alternative Z:** \$400,000
 - **Savings of Alternative Z:** \$450,000

$$\text{Net Savings for Z} = 450,000 - 400,000 = \mathbf{50,000}$$

Step 4: Compare Net Savings

- We will now compare the net savings of all alternatives to determine the best choice.
- **Net Savings:**
 - **Alternative X:** \$100,000
 - **Alternative Y:** \$100,000
 - **Alternative Z:** \$50,000

3. Final Answer

- **Answer:** The best alternatives based on net savings are Alternative X and Alternative Y, both yielding **\$100,000** in net savings. Alternative Z is less favorable with **\$50,000**.
- **Check:** Both Alternatives X and Y provide the same net savings, while Alternative Z is significantly lower.

Quality Assessment: good

Issues: - The problem statement is clear, but it could benefit from a more explicit definition of 'best choice' since multiple alternatives yield the same net savings. - The solution does not explicitly state that both Alternatives X and Y are equally viable choices, which may confuse readers.